NMAM INSTITUTE OF TECHNOLOGY, NITTE Off-Campus Centre of Nitte (Deemed to be University)
Department of Chemistry
CHY104-MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT

MULTIPLE CHOICE QUESTIONS
UNIT – I ELECTRODE & ENERGY SYSTEMS

1.	**In a galvanic cell,**			
	A)	Anode is positive terminal	B)	Cathode is negative terminal
	C)	Anode is negative terminal	C)	Reaction is non-spontaneous
Ans	**C)**	**Anode is negative terminal**		
2.	**An electrode is considered as a cathode when it has ……..**			
	A)	High reduction potential	B)	Low reduction potential
	C)	High oxidation potential	D)	Depends on the nature of metal
Ans	**A)**	**High reduction potential**		
3.	**A working concentration cell is made of two half cells having ……..**			
	A)	Identical electrolytes and identical concentrations	B)	Identical electrolytes but different concentrations
	C)	Different electrolytes and identical concentrations	D)	Different electrolytes and different concentrations
Ans	**B)**	**Identical electrolytes but different concentrations**		
4.	**Electrode potential of calomel electrode ……..**			
	A)	Varies directly with concentration of KCl	B)	Varies inversely with concentration of KCl
	C)	Is independent of concentration of KCl	D)	Depends only on the concentration of calomel
Ans	**B)**	**Varies inversely with concentration of KCl**		
5.	**When clubbed with copper electrode, calomel electrode acts as ……..**			
	A)	Anode	B)	Cathode
	C)	Either as anode or cathode	D)	Auxiliary electrode
Ans	**A)**	**Anode**		
6.	**A calomel electrode is an example for ……..**			
	A)	Reference electrode	B)	Ion selective electrode
	C)	Inert electrode	D)	Gas electrode
Ans	**A)**	**Reference electrode**		
7.	**The calomel electrode is represented as ……..**			
	A)	Hg | Hg_2Cl_2 (s) | KCl (saturated or standard)	B)	Ag | AgCl (s) | KCl (saturated)
	C)	Hg | $HgCl_2$ (s) | KCl (saturated or standard) |	D)	Ag | AgCl (s) | HCl (0.1M) | glass
Ans	**A)**	**Hg | Hg_2Cl_2 (s) | KCl (saturated or standard)**		
8.	**Which is the sparingly soluble salt present in calomel electrode?**			
	A)	HgCl	B)	Hg_2Cl_2
	C)	KCl	D)	$HgCl_2$
Ans	B)	**Hg_2Cl_2**		
9.	**EMF of electrolyte concentration cell is independent of ……..**			

	A)	Number of electrons involved	B)	Temperature
	C)	Standard electrode potential of electrode	D)	Electrolyte concentration
Ans	**C)**	Standard electrode potential of electrode		
10.	**The potential of the calomel electrode depends on the concentration of ……..**			
	A)	Hg	B)	Hg_2Cl_2
	C)	KOH	D)	KCl
Ans	**D)**	**KCl**		
11.	**What is the electrode potential of the calomel electrode with saturated KCl solution at 298K?**			
	A)	0.334 V	B)	0.281 V
	C)	0.242 V	D)	0.0591 V
Ans	**C)**	**0.242 V**		
12.	**The reference electrode that can be used to determine the potential of zinc electrode is ……..**			
	A)	Glass electrode	B)	Platinum electrode
	C)	Copper electrode	D)	Calomel electrode
Ans	**D)**	**Calomel electrode**		
13.	**A glass electrode is commonly used to measure ……..**			
	A)	Oxidation potential	B)	Reduction potential
	C)	**pH of a solution**	D)	Conductivity of a solution
Ans	**C)**	**pH of a solution**		
14.	**Glass electrode is sensitive to …… ions.**			
	A)	Sodium	B)	Hydrogen
	C)	Chloride	D)	Hydroxide
Ans	**B)**	**Hydrogen**		
15.	**Which of the following is an example for ion selective electrode?**			
	A)	Platinum electrode	B)	Gas electrode
	C)	Calomel electrode	D)	Glass electrode
Ans	**D)**	**Glass electrode**		
16.	**Identify which of the following statement/s does/do not hold true for glass electrode?** **(i) The glass membrane is made of a mixture of silicon, sodium and carbon oxides.** **(ii) The glass membrane exhibits electrical conductivity.** **(iii) The potential of the glass electrode is a combination of boundary potential, asymmetry potential and internal reference electrode potential.**			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (iii)	D)	Both (ii) and (iii)
Ans	**A)**	**Only (i)**		
17.	**Which of the following is an example for working concentration cell?**			
	A)	$Fe\|FeSO_4(0.10M)\|\|CuSO_4(0.10M)\|Cu$	B)	$Mg\|Mg^{2+}(aq)\|\|Cd^{2+}(aq)\|Cd$
	C)	$Ag\|AgNO_3(0.10M)\|\|AgNO_3(0.50M)\|Ag$	D)	$Fe\|Fe^{2+}(0.015M)\|\|Ag+(0.13M)\|Ag$
Ans	**C)**	**$Ag\|AgNO_3$ (0.10M) $\|\|AgNO_3$ (0.50M) $\|Ag$**		
18.	**Among the below given statements, which of them are true for concentration cell?** **(i) Electrode immersed in dilute solution behaves as anode.** **(ii) The Nernst equation of a concentration cell is independent of temperature and electrode.** **(iii) Concentration cells can be used to measure ion concentration in a solution.**			
	A)	(i), (ii) and (iii)	B)	(i) and (ii)
	C)	(ii) and (iii)	D)	(i) and (iii)

Ans	**D)**	**(i) and (iii)**		
19.	**The EMF of the concentration cell Ag(s)|Ag^+(0.05M) ||Ag^+(0.05M)|Ag at 298ºK is ……..**			
	A)	0.0591V	B)	0.02955V
	C)	0V	D)	-0.0591V
Ans	**C)**	**0V**		
20.	**The number of galvanic cells, each of 1.1 V, required to construct a 11 V battery is ……..**			
	A)	15	B)	22
	C)	11	D)	10
Ans	**D)**	**10**		

21.	**Which of the following is not a component of a battery?**			
	A)	Cathode	B)	Salt-bridge
	C)	Anode	D)	Electrolyte
Ans	**B)**	**Salt-bridge**		
22.	**Which of the following is not a feature of secondary batteries?**			
	A)	Rechargeable	B)	Long-lasting when charged
	C)	Non-rechargeable	D)	Can be used after recharging
Ans	**C)**	**Non-rechargeable**		
23.	**…….. is used as anode material in commercial lithium-ion batteries.**			
	A)	Lithium intercalated in Graphite layers	B)	Silicon
	C)	Lithiated metal oxides	D)	Platinum
Ans	**A)**	**Lithium intercalated in Graphite layers**		
24.	**The role of polypropylene separator in lithium-ion battery is to ……..**			
	A)	Block the flow of Ions	B)	Allow the free flow of ions
	C)	Allow the free flow of electrons	D)	Block the flow of electrons
Ans	**B)**	**Allow the free flow of ions**		
25.	**In lithium-ion battery, the electrolyte used is ……..**			
	A)	Polar solvent	B)	Non-polar solvent
	C)	Lithium salts dissolved in polar solvent	D)	Lithium salts dissolved in non-polar solvent
Ans	**D)**	**Lithium salts dissolved in non-polar solvent**		
26.	**Special properties of 'Li' metal that make it advantageous as an electrode material include ……..**			
	A)	Light weight	B)	Low electrical conductivity
	C)	High electrode potential	D)	Low cost
Ans	**A)**	**Light weight**		
27.	**The process during charging of a Li-ion battery involves ……..**			
	A)	Li^+ ions moving from cathode to anode		Li^+ ions moving from anode to cathode
	C)	H^+ moving from cathode to anode		H^+ moving from anode to cathode
Ans	**A)**	**Li^+ ions moving from cathode to anode**		
28.	**In flow batteries, the electro-active material is stored in ……..**			
	A)	Electrode	B)	Electrolyte
	C)	Separator	D)	Membrane
Ans	**B)**	**Electrolyte**		
29.	**Identify the incorrect statement for vanadium redox flow battery.**			
	A)	The redox reactions involve vanadium in different oxidation states.	B)	The electrolytes are stored externally and pumped through the cell.

	C)	The energy is stored in the electrode material.	D)	The electrolytes are rechargeable.
Ans	**C)**	**The energy is stored in the electrode material.**		
30.	**In a vanadium redox flow battery, the electrolyte contains** ……			
	A)	Vanadium ions of same oxidation states	B)	Vanadium ions in different oxidation states
	C)	Lithium ions	D)	Copper ions
Ans	**B)**	**Vanadium ions in different oxidation states**		
31.	**Which ion pair is used in the negative half-cell of a vanadium redox flow battery?**			
	A)	V^{3+}/V^{2+}	B)	V^{5+}/V^{4+}
	C)	Fe^{3+}/Fe^{2+}	D)	Cu^{2+}/Cu^{+}
Ans	**A)**	V^{3+}/V^{2+}		
32.	**In VRFBs, the battery energy and capacity depend on …….**			
	A)	Electrode material only	B)	The volume and concentration of electrolytes
	C)	The size of the membrane	D)	The temperature of operation
Ans	**B)**	**The volume and concentration of electrolytes**		
33.	**Photovoltaic cells convert …….. energy to …….. energy.**			
	A)	Chemical; Electrical	B)	Electrical; Chemical
	C)	Solar; Electrical	D)	Electrical; Solar
Ans	**C)**	**Solar; Electrical**		
34.	**The most commonly used semiconductor material in commercial solar cells is ……..**			
	A)	Gallium arsenide (GaAs)	B	Silicon (Si)
	C)	Cadmium sulfide (CdS)	C	Germanium (Ge)
Ans	**B)**	**Silicon (Si)**		
35.	**The working principle of PV cell is based on ……..**			
	A)	Ohm's law	B)	Nernst Equation
	C)	Newtons laws	D)	Photoelectric effect
Ans	**D)**	**Photoelectric effect**		
36.	**Which of the following statement/s is/are true?** **(i) In a PV cell, the p-doped silicon layer is exposed to sunlight.** **(ii) Depletion zone is responsible for separation of electrons and holes.**			
	A)	Only (i)	B)	Only (ii)
	C)	Both (i) and (ii)	D)	Neither (i) nor (ii)
Ans	**B)**	**Only (ii)**		
37.	**Which of the following property makes it easy to purify silane and chlorosilanes?**			
	A)	High density	B)	High boiling point
	C)	Volatility	D)	Magnetic property
Ans	**C)**	**Volatility**		
38.	**Which compound is formed when silane is pyrolyzed at 600–900°C?**			
	A)	SiO_2	B)	Silicon nitride
	C)	Trichlorosilane	D)	Polysilicon
Ans	**D)**	**Polysilicon**		

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